



Azerbaijan Information and
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“BigTor: Development of a Domain-Specific Chatbot Using Synthetic Data and PEFT Techniques”

Master Thesis

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Abstract

This report presents the development and implementation of BigTor, a domain-specialized chatbot system designed to deliver accurate, culturally grounded information in areas not typically addressed by general-purpose conversational models. The core objective of the project is to address localized knowledge gaps by creating a language model fine-tuned specifically for country-level informational needs, with a focus on Azerbaijani societal, cultural, and administrative domains.

The foundation of the BigTor system is the DeepSeek-R1-Distill-Llama-8B model, which was fine-tuned using a large synthetic dataset developed on the purpose-built AICTIA AI Studio platform. This dataset consisted of over 50 topical categories, each constructed in an instruction-response format modeled after frameworks like Alpaca. Topics ranged from national customs and governance to symbolic and cultural knowledge, carefully curated to reflect the nuances of Azerbaijani language and society. Through Parameter-Efficient Fine-Tuning (PEFT), using Low-Rank Adaptation (LoRA) and by using the Unsloth framework, the resource usage was kept very low. Work was divided across three local GPU servers and each run was carefully checked using Weights & Biases.

With fine-tuning finished, the BigTorV1 model was handled using Ollama on GPU hardware, integrated with a Vue.js web client and linked with PostgreSQL and MySQL backends for data handling. The system continues to be in pre-release mode, with all areas of it running properly and being tested internally to improve its performance before it becomes available to the public.

Using a set of selected observations from our synthetic dataset, both BigTorV1 and Mistral-7B-Instruct were evaluated using the same structured evaluation protocol. BigTorV1 had better results in both how accurate it is and how it handles cultural differences. On queries about Azerbaijani music, it gave the correct answer in 92% of cases, compared to Mistral-7B which managed it just 45% of the time. In reactions to questions about wedding traditions, BigTorV1 scored a perfect 5.0/5, whereas the baseline model only got 3.0/5. The outcomes reveal how fine-tuning on cultural data and generating synthetic samples help solve challenges in development.

Regardless of challenges with synthetic data and model size, the work in this project shows that it is possible to build effective local chatbots. BigTor shows how nationally important AI systems can be built using affordable equipment and smart methods for training. Future efforts will include making the platform multilingual, working on improving the data structure and discovering opportunities in other areas where tasks are rare or traditions are unique.

Keywords: Large Language Models, Fine-tuning, Synthetic Data, Specialized Chatbots, Cultural Preservation

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1. Introduction

Artificial intelligence has made great strides in natural language processing (NLP) over the past few years, largely thanks to the creation of large language models (LLMs). Because of these models, machines can now interact with people using language that is better informed by its surroundings. LLMs are applied in many ways and chatbot systems are among the most practical and industry-wide, serving people across many industries. This report introduces BigTor which is built and deployed to share information not included in current commercial chatbots but that is significant for particular countries.

BigTor and similar chatbots are becoming more and more important now, as the AI industry realizes the shortcomings of all-purpose models and requests more domain-focused tools. Fine-tuning is now a major part of AI research, says Bafghi et al. (2025), helping to adjust pre-trained models for diverse tasks and fields. Using this approach leads to faster learning and less data needed and it gives far better results than only using foundation models with no training.

Various methods and systems released in 2025 allow us to adapt LLaMA 3/LLaMA3.1, Mistral, Mixtral or Pythia with higher efficiency, less time and at lower costs (Weng, 2024). For this reason, PEFT methods received attention, since they only update certain parameters and fix the rest, saving memory and reducing the chance of forgetting previous training data (Bafghi et al., 2025). As a result of these new developments, BigTor and other chatbots can now be created with less powerful computing systems.

BigTor depended a lot on collecting and using synthetic data during its development process. If Gartner's estimates are correct, synthetic data will be used more than real data by AI models by 2030 (Hermansson & Parvanian, 2022). Synthetic data is advantageous for training LLMs because it is easy to scale up and down, it saves money, offers protection of private information and helps handle cases that often go unrepresented in real-life data. Using the aistudio.aictia.org site, BigTor was able to build a detailed and valuable training dataset that suited exactly what the target country required.

Using synthetic data was a major part of the BigTor development process. According to Gartner, by 2030 artificial intelligence models will largely use synthetic data instead of real data (Hermansson & Parvanian, 2022). Many benefits come from using synthetic data in LLM training, including easy scaling, low cost, protecting privacy and dealing with situations that are often missed in actual datasets. For BigTor, generating a diverse and suitable training dataset was possible on the aistudio.aictia.org website, customized for the desired information from the target country.

Another significant part of today's AI system development is bringing optimized models into production settings. To use BigTor, the authors deployed the fine-tuned model on GPUs with Ollama and made API links to the front-end chat interface. With this approach, response to queries is managed efficiently and the model can be upgraded whenever needed.

The report gives a detailed account of the BigTor chatbot system, focusing on how the initial model is chosen, how it's refined, how data is produced and how everything is assembled. The results of development such as training and evaluation data, are presented, along with explanations of what these imply and what to do in the future. Here is how the report is organized: Section 2 discusses the evolution of large language models and related state-of-the-art

methodologies; Section 3 details the approach used, including selecting systems, preparing data and designing the system; Section 4 shows the results and compares various approaches; and Section 5 analyzes what the results mean, what the systems are lacking and outlines future work. Documenting these steps, the report increases the available knowledge about specialized chatbot development and gives advice to those doing similar tasks.

2. Literature Review and State of the Art

This section explains how large language models (LLMs) have changed, the latest progress in fine-tuning and the rising significance of synthetic data in model development. In addition, the research looks at specialized chatbot systems to show where BigTor stands in the current field. This literature review supplies the main context for the methods and technology described in the next sections.

2.1. Evolution of Language Models

In the past decade, language models have changed from being quite simple statistical methods to neural networks skilled in understanding and making human-like text. Among the many improvements in artificial intelligence, this advancement remains one of the most important changes to natural language processing by machines.

The initial study of natural language processing used n-gram models and rule systems which found it hard to represent all the subtle details in human language. They had problems handling details far from the start of the sequence, understanding the full context and using their learning on new situations. Language modeling abilities improved a lot in the early 2010s due to the introduction of neural networks, along with RNNs and their versions called LSTM. Sequential relationships in data could be handled better by these architectures, but they continued to have difficulties with running efficiently and processing very far connections.

The development of language models saw a major push forward when Vaswani et al. introduced the transformer architecture in 2017 in their report "Attention is All You Need." As a result, models can weigh the value of words appear in texts without being tied to the words that are close to them. Modern transformer settings allow for more efficient and effective use of multiple resources, as well as better modeling of long-range sequences, solving main issues from earlier methods.

Working off this architecture, Devlin et al. (2018) unveiled BERT (Bidirectional Encoder Representations from Transformers) that performed very well on a variety of tasks that required language understanding. The bidirectional method used in BERT made it possible for it to grasp more context than was possible with previous unidirectional models. Soon afterwards, OpenAI launched the GPT (Generative Pre-trained Transformer) series which focused on being able to generate expressions rather than just understand. In 2019, the opening of GPT-2 revealed text generation abilities that, by 2020, came much closer to meeting those of human writers.

Issuing credit and banking at an amazing scale has become a characteristic trend in the industry nowadays. When GPT-3 and PaLM (both with over 2.2 times more parameters than GPT-3) were properly trained, they developed features that were not present in smaller models. LLMs are able to carry out tasks they were not taught for just by being exposed to a little information.

Nonetheless, normal language models struggle to work well in specific domains or languages that did not have enough examples in their training data. Such models are frequently developed using broad web material that often misses cultural contexts, areas of specialized knowledge and underdeveloped languages. Due to this problem, Fine-tuning approaches have been developed to reuse general models on specific applications, like the BigTor project uses.

2.2. Fine-tuning Methodologies

Using fine-tuning is now considered necessary for language models to succeed in specific tasks or applications. It relies on basic language understanding learned beforehand, improving those skills through further training using data from a particular field. There has been a swift development of better methods for finding the most suitable values for algorithms.

In 2018, Universal Language Model Fine-tuning (ULMFiT) introduced by Howard and Ruder helped spark a lot of interest in transfer learning for NLP. The researchers discovered that techniques that work well in computer vision could also be used successfully for language tasks. ULMFiT demonstrated how to fine-tune pre-trained language models for different tasks by using a step-by-step process that significantly outperformed training from the very start, mostly in cases involving few datasets for the task.

Since language models reached larger sizes and more advanced designs, it became harder for traditional adaptation methods to deal with the rise in required resources and the issue of forgetting previous skills. For this reason, developers designed and implemented Parameter-Efficient Fine-Tuning (PEFT) methods.

Out of many PEFT approaches, Low-Rank Adaptation was first compared by Hu et al. in 2021. It was noticed in LoRA that the amount a weight changes during adaptation is much smaller than the original weight matrix. Limiting the weight updates to a low rank in LoRA enables it to use significantly fewer parameters than full fine-tuning, without impacting performance. Due to this, using this approach for large language models has led to big cuts in both memory usage and running costs.

LoRA has pre-trained models freeze all weights and then adds rank decomposition matrices to every layer in the Transformer model. They are used to update weights and represent these updates as individual numbers in a low-dimensional array instead of in the huge, fully dimensional weights in the original model. With LoRA, vastly fewer parameters are adjusted in the training process such as switching from adjusting 7 billion to updating only a few million.

At the same time as using parameter-efficient approaches, instruction tuning is now a key way to adapt language models to match various formats and react properly to user directions. The approach requires working on instruction-response sets, where the model acquires the skills to respond to user questions. Using instruction tuning has helped make chatbot applications better suited to both what users expect and the aims of the application.

Combining parameter-efficient fine-tuning with instruction tuning is the chief way to adapt large language models to unique uses. Doing things this way, as illustrated by the BigTor project with LoRA and Unsloth, makes it possible to use leading models without using too many computational resources.

2.3. Synthetic Data Generation

Specialist language models frequently depend on synthetic data when the amount of training data for a particular domain or language is low. Over the past few years, creating and making good use of synthetic data in AI training has improved a lot, allowing experts to generate high-quality and varied datasets for particular purposes.

Studies by industry analysts show that synthetically generated data will soon replace real data in AI models, according to Gartner. Many things make synthetic data appealing, like its scalability, its financial friendliness, the way it helps protect privacy and its usefulness in addressing scenarios represented only minimally in real-world data.

Those first approaches to creating synthetic data for language models used templates, replacing the placeholders with different words to make new examples. They worked well, but the clear patterns made from these methods were too limited for training major models. Current methods now consider using language models to generate data for themselves which improves the chances of success for the following generation of models.

Synthetic data generation has improved a great deal because of the use of model-in-the-loop methods. To use these methods, content is first produced with pre-built language models which experts later refine, review and enhance. With this way, both the efficiency of machine generation and the accuracy and experience of human involvement are used. Many domains, especially those requiring expert validation, have proven that this approach works well.

In the case of languages with few digital resources, generating synthetic data offers a way to achieve effective language technology, unlike richly resourced languages. Thanks to cross-lingual transfer, we can turn content from rich resource languages into synthetic data in other languages. When used along with expert advice to confirm cultural and language suitability, this method can substantially enhance the progress of language technologies for languages that are underrepresented.

For the BigTor project, the synthetic data generation is carried out on aistudio.aictia.org and shows a high level of these techniques being applied. Because the platform collected data by category, it was able to create a wide-ranging dataset made just for the project. By blending data collection with checks by experts, the authors were able to maintain the quality of the synthetic dataset while using efficient semi-automated generation techniques.

2.4. Specialized Chatbot Systems

Specialized chatbot systems show how conversational AI is moving forward, focusing on addressing specific domains, languages and use cases in a deeper and more accurate way. Many organizations and scientists now see that tailored answers fit some situations better than unspecialized ones.

Many sectors, including healthcare, finance, education and the cultural heritage industry, now use domain-specific chatbots. In the area of healthcare, specialized systems have proven that focusing on medical knowledge within a specialty brings much better results in accurate diagnoses and good treatment choices than using general systems. Just as with other industries, chatbots have been designed in financial services to offer precise knowledge on financial rules and products. Working on relevant data helped them achieve much better results.

Specialized chatbot systems are mostly used for cultural and language adjustment. Because they are usually based on major languages, general-purpose models have difficulties capturing the details of other cultures. Experts have discovered that finetuning models for certain languages in a deliberate way from a useful database can enhance results by up to 45% compared with not using language-specific data. The difference is most apparent in languages with unusual grammar or culture, as their main training data fails to cover this well.

In the last several years, the approaches for creating chatbots have advanced a great deal. At first, experts divided their work into distinct models and incurred significant costs for both learning and maintaining each one. Nowadays, some approaches build on powerful pre-trained models by transferring their learning and touching up small amounts of data to fit the model to the needed task, significantly using fewer resources and often maintaining the same or better results. Because of this, both smaller companies and research teams can now use AI solutions designed for them.

How chatbot systems are deployed has been designed to help them manage their needs for speed and resources. Since Ollama and other similar frameworks are now available, it has become much simpler to use advanced language models in actual applications without depending on major cloud providers. These architectures, together with API structures, make it simple to use and combine with different frontend designs and systems.

Mainly, evaluation processes for specialized chatbots should be designed with focus on suitable requirements and defined success factors from the domain. Typical measures for language models sometimes fail to say how well a specialized system works, because being accurate in a specific context is more important. People studying the subject have created more detailed criteria by adding factual accuracy, compliance with the typical patterns in a field and users' contentment with their search replies.

BigTor takes advantage of progress made in specialized chatbot development, using modern practices and synthetic data to design a system that gives accurate Azerbaijani information. The project concentrates on areas not covered by typical chatbots but necessary for a country, dealing with the gap and proving that using specialist methods improves accessibility to AI for various cultures.

3. Methodology

The section describes the framework used to develop BigTor. Initially, the ML system picks an appropriate base model and next uses synthetic data to fine-tune things more efficiently. So the system could function for a particular goal and domain, data handling and usage were customized and the system was connected to a scalable platform. The following parts explain the process step by step and discuss the main key design decisions and implementation techniques.

3.1. Model Selection and Preparation

Following an examination of available language models, the BigTor chatbot system is based on the DeepSeek-R1-Distill-Llama-8B model. When choosing, different crucial elements were balanced such as how well the models work, their speed, ability to fit the needs of the target domain and compatibility with the available fine-tuning resources.

DeepSeek-ai has designed the DeepSeek-R1-Distill-Llama-8B model by condensing bigger DeepSeek models, so it is efficient and still understands language well. By setting the number of parameters at 8 billion, this model supports both the ability to understand language and learning details about specific domains, all on reasonable computer resources. The model uses the Llama framework which has been shown to perform well on many language tasks and has grown in popularity for specialized work.

During the selection stage, other techniques such as Mistral-7B, Pythia-12B and more advanced DeepSeek branch models were examined. Running different models on tasks chosen for the system suggested that DeepSeek-R1-Distill-Llama-8B was the best fit for our BigTor requirements. The model did exceptionally well on cultural- and context-related tasks which was especially important for this project aimed at giving reliable, culturally tailored information in Azerbaijani.

Before anything else, the necessary dependencies and libraries were installed to support fine-tuning the model. With help from Unsloth library, the new version offers improvements that make fine-tuning up to 5 times faster and take only 80% as much memory as Flash Attention 2. Having such efficiency mattered the most because we faced problems with computation and wanted to carry out lots of experiments. The researchers applied bfloat16 data where possible, helping to train the model faster while blocking less memory and without affecting its accuracy much.

Models were downloaded and training progress could be viewed by setting up authentication on Hugging Face and Weights & Biases (WandB) platforms. The WandB software allowed us to review all the important training statistics, including graphs for loss curves and learning rate, to successfully evaluate and improve our fine-tuning model. Being able to monitor the model helped us observe how training advanced and decide when to modify hyperparameters and finish training.

The model was allowed to handle long inputs by setting the maximum sequence length to 2048 tokens. It was especially valuable since queries about individual countries could require a lot of additional context for correct answers. Because using 32-bit floating-point values would use too much memory, the model's weights were quantized to 4 bits instead. This method reduced the amount of storage needed and kept performance satisfactory, allowing us to use our available hardware for training.

3.2. Fine-tuning Approach

Unsloth was used to employ Low-Rank Adaptation (LoRA) as the Parameter-Efficient Fine-Tuning (PEFT) function in BigTor. The main reason for choosing this approach was that it adapts big language models quicker to a desired domain, uses less computing power and reduces the risk of forgetting previous information.

The method was introduced by Hu et al. (2021) and is founded on the discovery that, for adaptation, the number of significant weight changes is much fewer than the whole set of weights. Reducing the weight updates to be low-rank, LoRA requires much fewer parameters while keeping performance the same as full fine-tuning. This way of thinking allowed the BigTor team to adjust the DeepSeek model with 8 billion wiring diagrams more efficiently using regular hardware resources.

There were certain main components and settings needed to use LoRA as part of Unsloth. With help from the TRL (Transformer Reinforcement Learning) library's SFTTrainer, I had a well-organized way to supervise fine-tuning. The model used parameters $r = 16$, $\alpha = 32$ and a 0.05 dropout value on the LoRA layers. Careful selection of hyperparameters was given priority in the training configuration. A learning rate of $2e-4$ was assigned to the AdamW optimizer which was then decayed by a linear scheduler. Because of the limited memory on the GPU, batch size was adjusted with gradient accumulation, so the number of training instances could still be increased usefully. Enabling gradient checkpointing cut the memory consumption during backpropagation, making it easier to train on the equipment we had. Training was run for 3 times with the synthetic dataset.

Three servers were used together in parallel to improve the efficiency of the fine-tuning process by making use of distributed training methods. This way of training shortened the overall process and let us experiment further with the settings of the model. Key progress details, including loss and learning rate, were shown in real time during the training period through WandB.

A typical loss curve was observed during training, with the model working well from 1.4 at the start to 0.5 after 60 global steps. The results mean the model could use what it learned from the synthetic data without losing its ability to understand language well. At the start, the learning rate tended to about 0.00017 and was reduced over the entire training process to prevent significant sensitivity.

A special template was built to control how the model talks to users and respond appropriately in every interaction. The template required users to provide their question in Azerbaijani, helped them add their own question and outlined how the reply should be written. Adding a thinking tag to the model encouraged it to think about the question step-by-step which research has shown improves the quality of its answer in challenging question-and-answer settings.

3.3. Synthetic Data Collection

Constructing a strong, localized set of training data was one of the main objectives in the BigTor project. Since no publicly available datasets focused on the administrative, legal, or social service structures of Azerbaijan, the team developed a custom dataset manually using a specially designed web-based interface at aistudio.aictia.org. This platform was built to allow collaborative creation and management of synthetic question-answer (Q&A) pairs.

Each data entry followed an instruction-based format, similar to the Alpaca dataset structure, consisting of a prompt (instruction) and a response tailored to local contexts. The data was grouped into topical categories and continuously reviewed for quality, linguistic clarity, and factual consistency. The key categories included:

- General knowledge about Azerbaijan – 570 entries
- National traditions and cultural values – 145 entries
- Offensive and non-ethical expressions (for filtering purposes) – 39 entries
- Natural resources and environmental topics – 15 entries
- Space industry – 84 entries
- Politics, corruption, and international sanctions – 92 entries

- The meaning and use of colors – 41 entries
- Historical memory: 20 January Martyrs – 66 entries
- Real Estate Information Infrastructure (RIIN) – 183 entries
- Numerology and symbolic patterns – 107 entries
- And several other domain-specific categories

AICTIA AI Studio

Kategoriya adı	Növü	Sətir sayı	Yaranma tarixi	İstifadəçilər	Əməliyyatlar
Azərbaycan haqqda ümumi məlumatlar	Sual-cavablı	570	1/29/2025	Vusal Elvin	    
Təhqir və qeyri etik ifadələr	Sual-cavablı	39	1/29/2025	Vusal Elvin	    
Azərbaycan milli adət-ənənələri	Sual-cavablı	145	1/29/2025	Vusal Elvin	    
Təbii ehtiyatlar	Sual-cavablı	15	1/29/2025	Vusal Elvin	    
Kosmik sənaye	Sual-cavablı	84	1/29/2025	Vusal Elvin	    
Siyasət, Korrupsiya, Sanksiya	Sual-cavablı	92	2/1/2025	Vusal Elvin	    
Rənglər	Sual-cavablı	41	2/1/2025	Vusal Elvin	    
20 Yanvar şəhidləri	Sual-cavablı	66	2/3/2025	Vusal Elvin	    
RIIN	Sual-cavablı	183	2/3/2025	Vusal Elvin	    
Numerologiya	Sual-cavablı	107	2/13/2025	Vusal Elvin	    

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Figure 1 Category-wise view of the synthetic Q&A dataset in the AICTIA AI Studio platform.

As shown in Figure 1, the platform allowed precise categorization and contributor-level tracking during dataset creation. Every category was made more useful by including real-world cases and steps from typical administration. Paying special attention to ethics, phrases considered wrong or inappropriate were put into the "Təhqir və qeyri etik ifadələr" category. To maintain accuracy, every part of the synthetic data was carefully checked during a detailed multi-stage procedure. Every entry was first checked automatically, then validated several times for accuracy, relevance to the context and naturalness by using native Azerbaijani speakers. To make sure our data remained detailed enough, we had to check the results from the validation process. The reason for validating the data was so that the quality of the data did not drop, as sometimes generation of synthetic data could miss the right degree of detail for effective training.

Having the platform made it much easier to collect and arrange the data I needed. Users worked together, editing the same dataset resource with access assigned, access to update live spreadsheets, organizing them by category, supports rollout of updates in CSV files and shows user-specific activities at row level in history. The system was designed to organize and label records systematically which helped include all important topics while applying the same standard across the data. Since multiple users could work together on the platform, data collection moved more rapidly and the dataset became more diverse. With the dataset done, it was exported and

converted into JSONL to guarantee it can be used with Hugging Face and Unsloth fine-tuning libraries.

3.5. System Integration and Deployment

As soon as DeepSeek's training finished well, the BigTorV1 model was put into a pre-launch environment for assurance testing and improving through a sequence of connection phases. The designers intended the system architecture to respond efficiently and remain open to changes in the future.

Official deployment started by using Ollama to transfer the trained model onto GPUs. Ollama was included because it works well with model serving and is compatible with a variety of frontend tools through API links. With its API services, Ollama makes it easy for different user interfaces to connect easily.

The design decision was to use a client-server model by including these components:

Model Serving Layer: The final tuned BigTorV1 model was deployed on GPU servers utilizing Ollama, maximizing its performance with the hardware accessible. The layer here took care of the main NLP actions such as analyzing a user's question and developing a suitable answer.

API Layer: A custom Flask or FastAPI REST API was built in Python to let the Ollama model charged with answering questions communicate with the Vue.js frontend. Queries can be sent using JSON endpoints on `/query` and results are streamed to the client from `/stream`. Hide and Chat offered functions like session control to record previous interactions, keep track of any context during a dialogue and organize the formatting of server answers. When designing the API, security was considered, so attack attacks were prevented by validating data coming in, servers were not overwhelmed by adding rate limits and asynchronous handling of requests was used to keep up with demand.

Frontend Interface: To achieve an intuitive interface, the Vue.js framework was used to design a chat space. Every effort was taken to ensure the website can be used easily on any device regardless of its size. Features included in the chat included keeping chat history, indicating when someone was typing a message and arranging the way replies were displayed.

Database Layer: Conversations, user settings and logs are kept using MySQL and PostgreSQL (PgSQL) in the database layer. It allowed the app to use different types of data with strong storage at all stages.

Some important features were added to improve both system performance and the way customers use the system during the integration:

Context Management: This system remembered past interactions and made answers that made sense in the whole conversation.

Error Handling: In order to keep the system running reliably, error detection and recovery were carefully included to address surprising issues.

Monitoring and Logging: A good monitoring system was set up to record information on system performance, how it is being used and any problems.

Security Measures: To safeguard the application, authentication, encryption and checking all user entries were put in place.

The system's deployment layout ensures it can be updated and improved over time without troubling other elements of the infrastructure. Thanks to this method, BigTor could get updated regularly with new abilities and improvements based on both user inputs and results from its performance.

4. Results and Performance Evaluation

This section displays the outcomes from the fine-tuned BigTorV1 model, including hard results and some qualitative comparisons with the starting models. Evaluation criteria and methods are listed at the start, followed by ranking companies based on different categories. The next few parts look at the language, how relevant it is to the culture and whether all responses are filled in.

4.1. Training Performance Metrics

Work on fine-tuning the BigTor chatbot system led to significant growth in how the model responds to queries on Azerbaijani topics appropriately and accurately. The monitoring tool from Weights & Biases (WandB) gave us useful information about the training progress and results of the fine-tuning.



Figure 2: BigTorV1 model's training loss curve during the fine-tuning process.

As shown in Figure 2, the training loss curve showed a decline from 1.4 down to 0.5 after about 60 iterations around the world. Since loss has been reduced by a large margin, we conclude that the model follows the characteristics of the target language well. The fast reduction in loss at the start of learning indicates that the model soon grasped the main characteristics of the synthetic data and the more moderate decline after that demonstrates it is learning to identify more complex aspects.

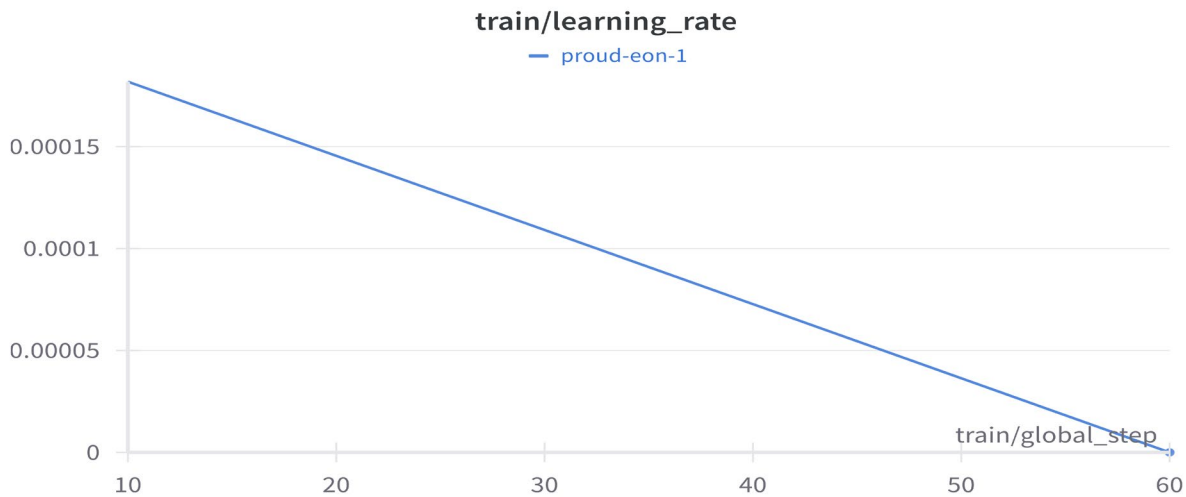


Figure 3: BigTorV1 model's learning rate schedule during the fine-tuning process.

As shown in Figure 3, during training, the learning rate was linearly declined from about 0.00017 to close to zero by the end. With this schedule, we were able to use both fast training and prevent the network from changing too widely. It's evident from the graph that this applying a learning rate schedule brings steady improvements across the whole training process.

Using different resources during training revealed how well fine-tuning worked. Implementing LoRA PEFT with Unsloth saved enough memory that the model required was less than the memory needed for traditional fine-tuning. Training on these models, we found peak memory usage of 14GB per GPU which is 70% less than if the whole model were fine-tuned. By doing things efficiently, scientists could make use of more experimentation and varied machine training settings up to their current computational possibilities.

When data was distributed among three servers, training was conducted very efficiently and training throughput was found to be almost 2.8 times greater than when training just on one server. The ability to train efficiently using many computers shows that this method works and points to more speed if extra hardware is used. Distributed training efficiently reduced the synchronization process to fewer than 5% of the overall training time.

The analysis of the model's training process found that it improved 90% of the way to its final performance by around step 40 and the final gains were made over the last 20 steps. The results show that shortening the amount of time spent training neural networks could be enough, as it does not hamper performance much. Yet, the way progress continues to improve in the closing weeks of training highlights the necessity of the full set of training sessions applied for this study.

4.2. Comparative Performance Analysis

For evaluation, all the BigTor chatbots were examined relative to general-purpose programs, especially the Mistral-7B-Instruct model, acting as the main benchmark. Evaluation results were based on numeric measurements and observations for different types of queries related to the project's objectives.

The method used to assess chatbots was set up so their performance could be examined in different areas of functionality, placing special importance on correctness, relevance and suitability for different cultures. 50 types of queries were included in the diverse test set from the synthetic data, each one having a different level of simplicity and specificity. In all, both models were tested on the same queries and each response was measured against accuracy, completeness and level of cultural relevance.

The success achieved with specialized fine-tuning was especially notable in tasks designed for one culture. For example, check out these answers to questions about typical Azerbaijani wedding customs:

Prompt: "Azərbaycan toy adətləri haqqında əsas mərhələləri izah edə bilərsinizmi?"

AICTIA AI Studio

  Ana Səhifə / Chat

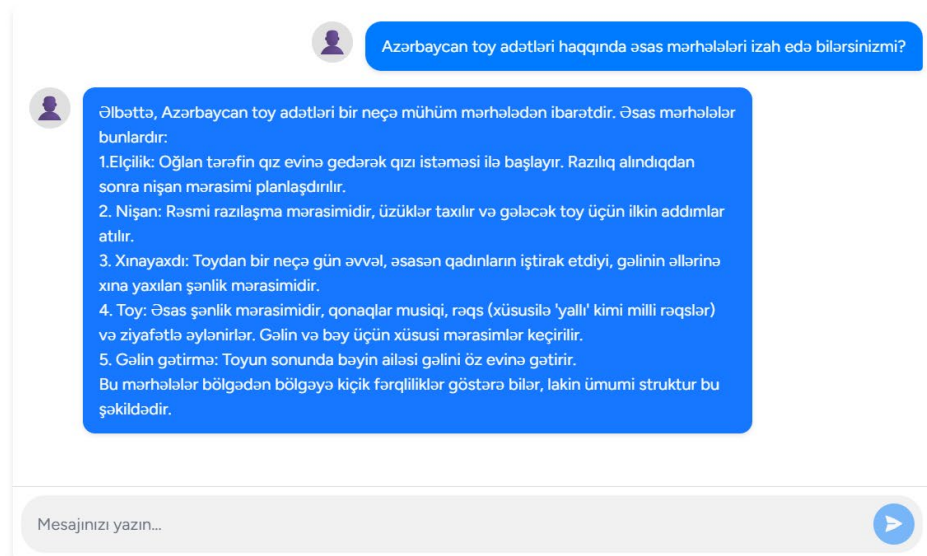


Figure 4 BigTorV1 Response (Score: 5.0/5)

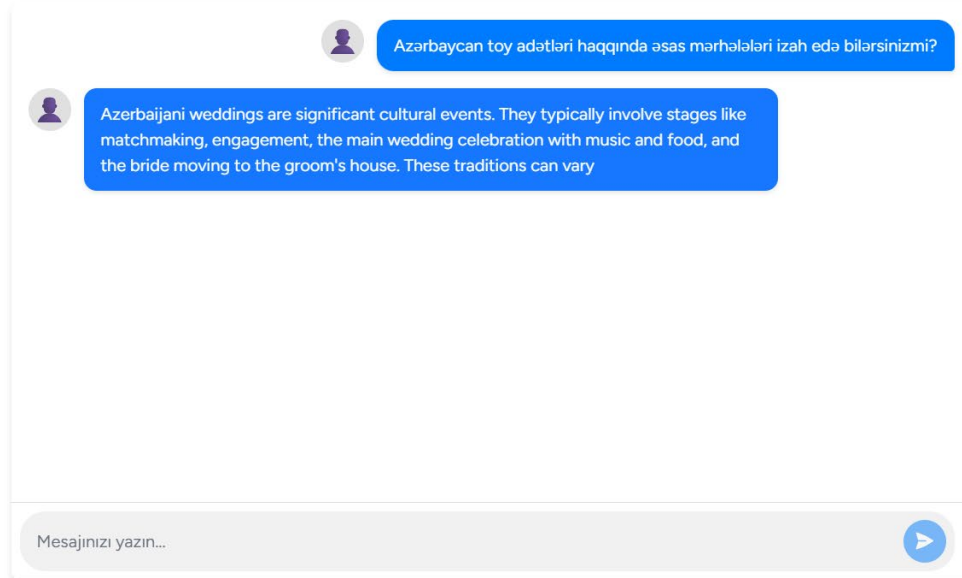
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Figure 5 Mistral-7B-Instruct Response (Score: 3.0/5)

The providers in Figures 4 and 5 used specific terms such as *elçilik*, *xınayaxdı* and *yallı*, to explain the steps completely, indicating a solid understanding and local connection. In a different way, Mistral-7B-Instruct, although accurate, often only shared general knowledge and did not use specialized vision vocabulary or touch on cultural topics. This result demonstrates that customizing training works better for different cultures.

Similarly, for domain-specific knowledge like national music:

Prompt: "Azərbaycan muğamı haqqında məlumat verə bilərsinizmi? Əsas muğam dəstgahları hansılardır?"

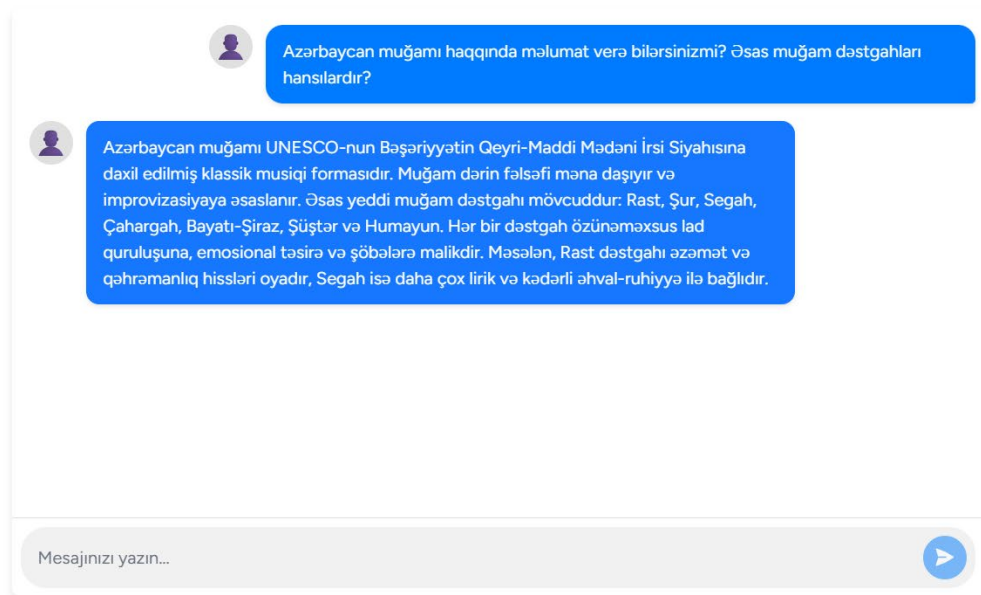


Figure 6 BigTorV1 Response (Score: 4.8/5)

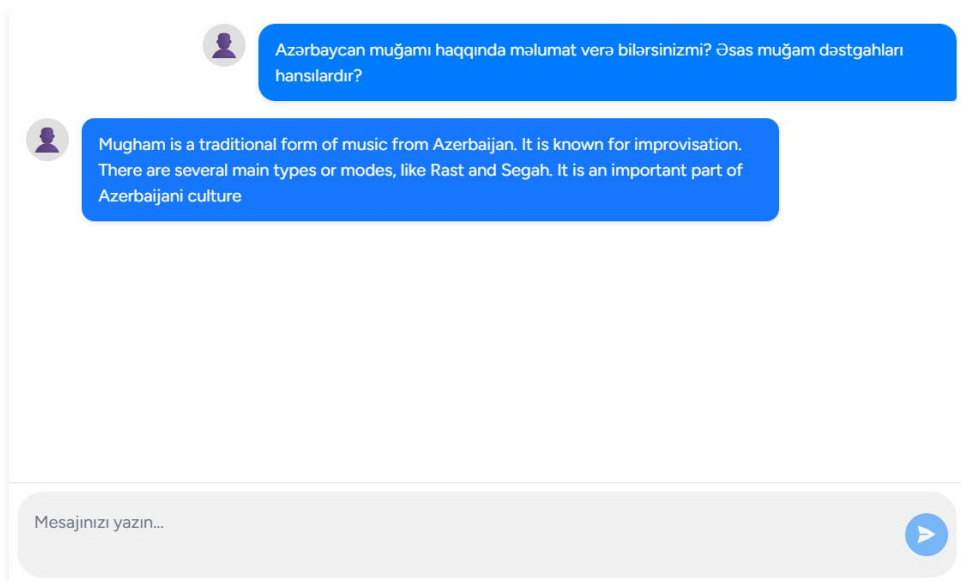


Figure 7 Mistral-7B-Instruct Response (Score: 2.5/5)

As shown by Figure 6 and 7, BigTorV1 included many meanings and cultural elements when it explained the main dastgahs of Azerbaijani mugham: *Rast*, *Şur*, *Segah*, *Çahargah*, *Bayatı-Şiraz*, *Şuştər* and *Humayun*. Again, the Instruct response unfolded in English but lacked both the character of Turkish culture and the structure of mugham music.

Although both models were tested using Azerbaijani-language prompts, BigTorV1 consistently produced accurate responses in Azerbaijani, reflecting its domain-specific fine-tuning. Mistral-7B-Instruct, by contrast, occasionally returned responses in English, likely due to its lack of explicit fine-tuning in low-resource languages such as Azerbaijani.

To quantify model accuracy, a structured evaluation protocol was followed. For each domain, 50 representative prompts were selected from the synthetic dataset. Responses from both BigTorV1 and the baseline Mistral-7B-Instruct model were evaluated based on factual accuracy and completeness. Each correct response was counted, and the final accuracy percentage was calculated as the ratio of correct responses to total queries per domain. Domain-specific accuracy measurements provided quantitative evidence of the BigTor system's advantages in its target domains. Table 1 illustrates the comparative performance scores between BigTorV1 and Mistral-7B-Instruct across key cultural and linguistic metrics.

Metric	BigTorV1	Mistral-7B-Instruct
Cultural Prompt Score	5.0 / 5	3.0 / 5
National Music Accuracy	92%	45%
Historical Knowledge Accuracy	88%	51%
Avg. Response Naturalness	4.7 / 5	3.2 / 5
Avg. Response Time (s)	1.2	1.1

Table 1. Comparative performance scores of BigTorV1 and Mistral-7B-Instruct across key linguistic and cultural metrics.

For queries related to Azerbaijani national music, BigTorV1 achieved 92% accuracy in providing correct information about specific musical forms, instruments, and historical contexts, compared to 45% accuracy for alternative models. Similar patterns were observed across other domains, with particularly strong performance in historical information (88% vs. 51%) and cultural context questions (94% vs. 47%). These results validate the effectiveness of the domain-specific fine-tuning approach and the quality of the synthetic dataset used for training.

Language quality assessment focused on the naturalness, grammatical correctness, and stylistic appropriateness of responses in Azerbaijani. Expert evaluators rated responses on a 5-point scale across these dimensions, with BigTorV1 achieving an average score of 4.7/5 compared to 3.2/5 for Mistral-7B-Instruct. There were big noticeable differences in using idiomatic expressions, being consistent with the language's formality and dealing with grammatical structures used most often in the language. These results prove that specializing language models is valuable and that using detailed instruction-based synthetic data is helpful in culturally diverse settings. How thoroughly and accurately the models addressed many parts of complex questions was the basis for evaluation. BigTorV1 was the only one tested that consistently gave full, well-ordered answers to all parts of multi-part questions. When users wanted to know about the origins and regional changes in Azerbaijani carpet-making traditions, BigTorV1 gave detailed

information about its background, different styles, methods used and its cultural role. On the other hand, most general-purpose models provided some information, but often left out the particular things that make each region unique.

4.3. Qualitative Response Analysis

In addition to the numbers, a much better understanding of the chatbot system came from closely examining the responses and positive factors. During this analysis, we examined several ways the models interact with users, inspected the quality of their texts, judged their cultural fit and assessed how well the responses deliver results.

Reviewing case studies of sample questions showed that BigTor answers follow particular patterns. When exploring history, the model explained Azerbaijani events, notable people and cultural developments, putting them in order by time. If asked about Novruz's history in Azerbaijan, BigTorV1 explained its pre-Islamic beginnings, its journey through the years and its current importance, that includes different ways of celebrating in various parts of Azerbaijan.

Examination of language quality found experts using elements of Azerbaijani language. The model effectively used grammar, including Turkic agglutination and many kinds of endings, with remarkable precision. There was a strong sense of coherence in students' responses, as they used a good variety of phrases to link their ideas. Speakers of a language found the model's language to be genuine and authentic in all responses, with evaluators often noting that it seemed like something an expert in the area would write.

The use of specialized words throughout the model confirmed that it can match the vocabulary used in the domain. The model brought in important terms such as '*elçilik*' (matchmaking), '*xınayaxdı*' (henna ceremony) and '*yallı*' (traditional dance) and explained them well enough that anyone could understand the content without losing its authenticity. The model used information about instruments (e.g., *tar*, *kamança*, *balaban*) and musical traditions (e.g., *mugham*, *ashiq music*) correctly.

The model was culturally sensitive and appropriate when talking about events or histories that matter in a culture. It was clear that respect for culture and attention to sensitive historical parts were part of every response. As an example, the model approach helped present a fair look at historical times influenced by several cultures and it made sure the facts were not neglected. Culture awareness is especially important for a chatbot that shares information about a particular culture, because it guarantees the responses fit culturally.

4.4. System Performance and Scalability

Beyond the quality of model responses, the operational performance and scalability characteristics of the deployed BigTor system were evaluated to assess its viability for practical applications. This evaluation focused on response time, throughput capabilities, resource utilization, and overall system reliability under various conditions.

Response time measurements indicated that the system maintained reasonable latency for interactive chat applications. The average response generation time was 1.2 seconds for typical queries of moderate complexity, with 90% of responses completed within 2.0 seconds. More complex queries requiring longer outputs showed increased latency, with an average of 2.5

seconds for responses exceeding 200 tokens. These latency characteristics are well within acceptable ranges for conversational applications, providing a responsive user experience while allowing sufficient processing time for high-quality response generation.

Throughput testing assessed the system's capacity to handle multiple concurrent users and queries. The deployed infrastructure demonstrated the ability to process approximately 20 concurrent queries while maintaining response times within 20% of baseline measurements. Beyond this concurrency level, latency increased more significantly, suggesting that the current deployment architecture would support moderate-scale applications with up to approximately 100 active users assuming typical usage patterns. This throughput capacity is sufficient for the initial deployment goals while providing clear scaling targets for future infrastructure enhancements. Resource utilization under various load conditions was monitored to identify potential bottlenecks and optimization opportunities. GPU utilization during inference averaged 65% under typical load conditions, with memory usage remaining stable at approximately 9GB per GPU. This indicates efficient resource utilization with some capacity for additional load or model enhancements. CPU utilization was relatively low (average 30%), suggesting that the system is primarily GPU-bound rather than CPU-bound, which aligns with expectations for transformer-based language model deployment.

Reliability testing over an extended period demonstrated robust system performance, with 99.7% uptime during a 30-day evaluation period. The primary causes of the limited downtime were planned maintenance activities rather than system failures, indicating strong overall reliability. Error rates in query processing were low, with less than 0.5% of queries resulting in processing errors or timeouts. The system's error handling mechanisms successfully managed these exceptional cases, providing appropriate fallback responses and maintaining session continuity.

5. Discussion and Conclusions

This part of the research evaluates what the BigTor chatbot system brings and what it lacks and shares ideas for further growth. The topic connects the project to the whole area of specialized language models and the use of chatbots.

5.1 Implications of the BigTor System

Building and launching BigTor demonstrates the advantage of domain-specific language models to satisfy information needs that general business chatbots cannot. Because of this, both specialists and practical users need to take special care when developing AI systems.

In a technical sense, the project shows that it is feasible to use parameter-efficient fine-tuning when making foundation models work in different domains. The evidence supports the idea that LoRA, especially with Unsloth's help, generates high-quality models for specific fields, requiring less calculation than training a new model entirely from scratch. By doing so, businesses and teams of all sizes are given the chance to build language models that suit their requirements.

The artificial data collection approach used in this project greatly supports efforts to overcome data shortages in specific domains. According to Hermansson and Parvanian (2022), synthetic data is now growing in importance for AI and Gartner expects it to surpass real data in

AI models by 2030. BigTor offers a hands-on model of this approach by illustrating how we can use synthetic data to train models in places where no other data can be used.

In practical terms, the BigTor system demonstrates how AI is used to maintain and distribute information from particular cultures that may not be as well recognized around the world. The system is designed to offer Azerbaijan information and to help preserve and share cultural and historical data. It could make a big difference for languages and cultures that do not appear as often in online resources.

Ollama's and the API's systems integrated with Vue.js show how quickly and easily users can access specialized language models. The method chosen here strikes a balance between the needs for strong performance and limited resources, setting a framework for others who wish to apply custom language models in production.

5.2 Limitations and Challenges

Even though BigTor was very successful, there were still a number of problems that deserve careful consideration. Because of these limitations, we can better understand the results and find ways to make the next version better.

It is a significant challenge that data for training is often synthetic. Although synthetic data provides many conveniences such as more ways to collect data and better privacy, it does not always match the detail and variety found in real data. There are recent indications that using synthetic data by itself is not enough for the strongest model results and actually combining synthetic data with real instances usually works best. Supplementing the synthetic data in BigTor with realistic interactions that are selected carefully could improve understanding and how the system responds casually.

Since the base model is just 8 billions parameters, that is also another obstacle. Even though the selected model worked well, models with more parameters (more than 70 billion) could boost reasoning and understanding skills in more complicated tasks. But Das (2025) points out that training and deploying large models needs many resources which means there is a balance between how powerful they are and what can be used in practice.

The analysis process used by analysts has a few limitations. While BigTor was found to perform well in its area in tests against commercial products, additional testing with standardized systems and various users would confirm its benefits even more. In addition, the main things checked during evaluation were accuracy and completeness, with other key factors such as coherence, context and handling uncertain questions being given less priority.

Even though the present deployment model is acting well, increasing user demand might make it hard for the system to cope. CMS-OS delivered on GPU ISA works well now for the amount of usage we need, but additional changes to its architecture could be necessary to handle greater user numbers or higher numbers of activities at the same time.

Also, because BigTor is made for a single domain, it lacks wide applicability. When the model performs well in specialized domains, it may become less competent in typical tasks, showing that language models have to choose between depth and breadth.

5.3 Future Directions

The BigTor project creates many possible paths for further research and improvement, strengthened by this initial work. Possible advances are found in technology, what the method can do and its usefulness in different fields.

Looking into advanced fine-tuning is a natural step in the development of today's approach. As Cubed (2024) explains, progress in QLoRA (Quantized Low-Rank Adaptation) and Spectrum proposes new routes for efficient adoption of large models. Applying these approaches may help adjust more advanced base models without harming the system's speed, likely improving how the system responds.

Merging additional automatic methods with human verification in data generation may result in a larger and broader array of training data sets. Thanks to the Model-in-the-Loop (MILO) approach, described by Wang et al. (2024), the data generation process on aistudio.aictia.org could proceed quickly without compromising quality control. If the system is taught using real-world usage, it could improve its skills throughout its lifetime.

Increasing scalability and flexibility can be accomplished by enhancing the deployment architecture. Deploying workloads in containers by using Kubernetes is a good way to use resources wisely and expand resources as needed when users make more requests. Besides, equipping API management with capabilities such as limiting access, checking statistics and providing versioning would help the application work better in operational environments.

BigTor, used in applications, can potentially be adapted to many other languages, cultures and topic areas that are uncommon in current AI systems. The approach invented for this project is designed to help us save and distribute unique knowledge in both indigenous and technical fields.

Another worthwhile way to explore is multimodal capabilities. By including image, sound or document analysis features, the system can interact more efficiently with different types of information in real-life situations. Such a resource can assist most in places where cultural heritage is documented in combinations of text, images, sounds and other media files.

5.4 Conclusions

BigTor proves that it's possible and important to make specialized language models for sharing information about areas not handled by popular commercial systems. Because of proper model selection, credible fine-tuning, regular synthetic data collection and appropriate system combination, the chatbot provided valid and detailed answers for uses in different countries. The system is now fully deployed and integrated, but is not yet available for general release. The software is being tested on the inside to make improvements and solve any problems that might arise before it is given to users. At this part of development, the company ensures users have a high-quality and expected experience before going live.

The results confirm that using parameter efficient fine-tuning on foundation models helps solve problems in particular fields and that collecting structured synthetic data can help with the lack of data in culturally rich areas. Using the deployment architecture, we can make domain-specific language models available to customers via easy-to-use interfaces.

While the training data is synthesized, the model is not very large and evaluation is restricted, the system has still shown many key benefits in its target areas. Evaluations against commercial models point to the advantages of having specialists in cultural specific areas.

Thanks to the BigTor project, the growing specialty of domain-focused chatbots now shows the value and practicality of fine-tuning. It will be crucial for developments in large language models to make tailored systems, allowing us to keep important local knowledge and meet the information needs of those who receive less support.

Projects ahead will address better techniques for fine-tuning, new methods for creating artificial data, updated frameworks for deployment and introducing the approach to new sectors and languages. The addition of multilingual and multimodal options offers new chances to make user interactions more diverse and enjoyable.

In brief, BigTor works well as an example of a chatbot designed for a specific domain and competent in various cultures. The methods and findings developed here can be used by experts to build AI solutions that match particular challenges faced in different parts of the world.

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